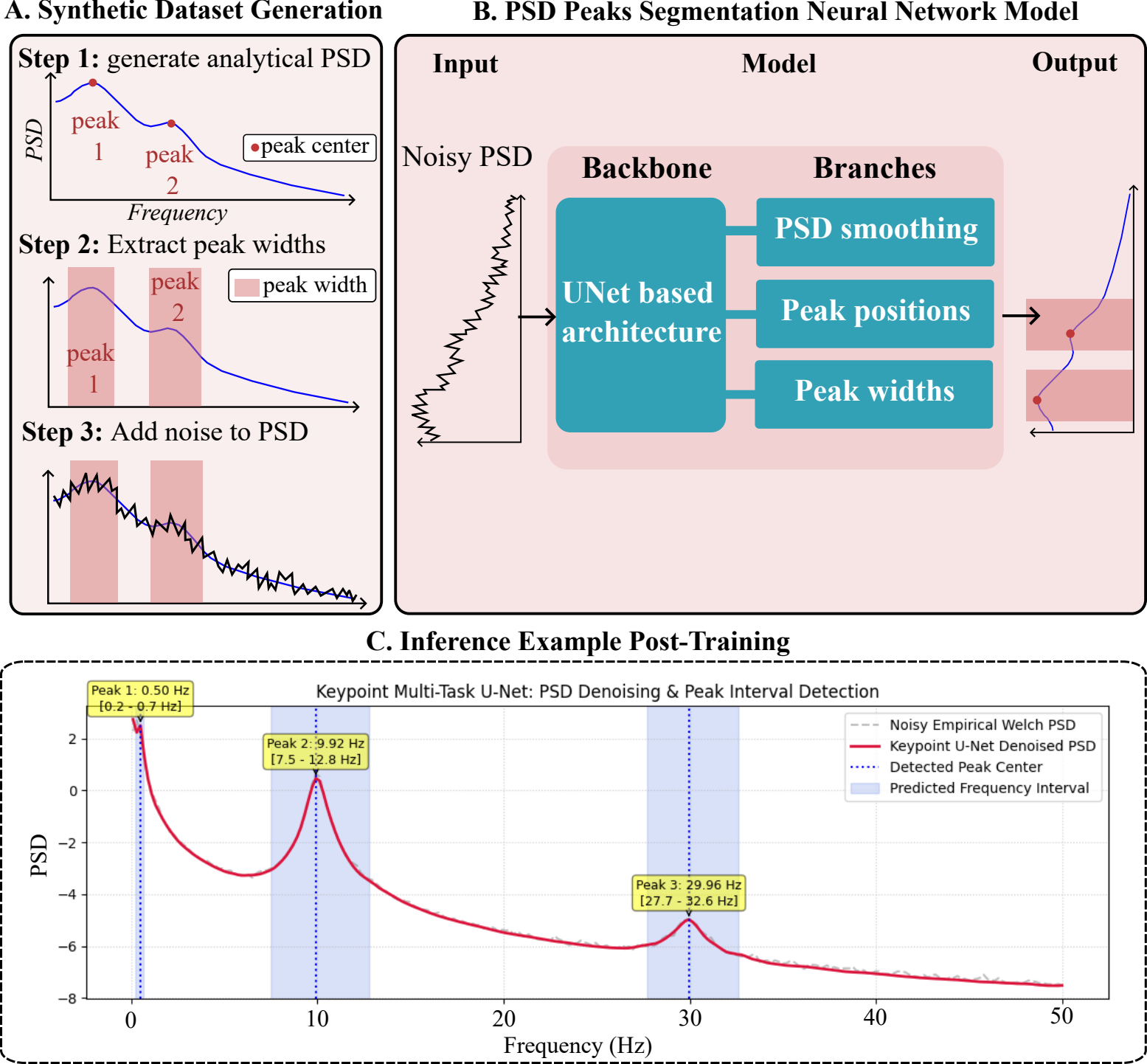

A. Synthetic Dataset Generation
Step 1: generate analytical PSD
peak 1
peak 2
peak center
PSD
Frequency
Step 2: Extract peak widths
peak width
peak 1
peak 2
Step 3: Add noise to PSD
B. PSD Peaks Segmentation Neural Network Model
Input
Model
Output
Noisy PSD
Backbone
Branches
UNet based architecture
PSD smoothing
Peak positions
Peak widths
C. Inference Example Post-Training
Keypoint Multi-Task U-Net: PSD Denoising & Peak Interval Detection
Peak 1: 0.50 Hz [0.2 - 0.7 Hz]
Peak 2: 9.92 Hz [7.5 - 12.8 Hz]
Peak 3: 29.96 Hz [27.7 - 32.6 Hz]
Noisy Empirical Welch PSD
Keypoint U-Net Denoised PSD
Detected Peak Center
Predicted Frequency Interval
PSD
Frequency (Hz)